Replication of "Measuring Growth from Outer Space"

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Introduction

GDP is among the most important measures of economic performance, but estimates of GDP are often quite uncertain. In the Penn World Tables, one of the standard compilations of cross-country GDP, for example, all 43 countries in Sub-Saharan Africa have margins of error of at least 20 percent, and some countries, such as Myanmar, have no data. Additionally, GDP data are not consistently available on a subnational level, limiting economists' ability to empirically investigate intranational variation in growth. Henderson et al. (2012) constructed a statistical framework for the use of satellite data on "night lights" as a measure of economic growth. Because the measurement error in night lights and official economic data are likely uncorrelated, the night lights proxy measure can be combined with official measurements of GDP to improve estimates of economic growth.

In this paper, I replicate Henderson et al.'s main analyses. My analysis was implemented in the R programming language (R Core Team 2026). The source code is available on Github, along with instructions for reproducing the results. The code is structured to facilitate extending the analysis to periods outside the original study's sample, although such analyses are outside the scope of the project.

Data

Since 1992, satellites from the U.S. Air Force's Defense Meteorological Satellite Program have measured the intensity of Earth-based lights using their Operational Linescan System. The raw data from DMSP-OLS are processed by the National Geophysical Data Center by removing sources of intense natural light such as forest fires and aurora borealis, leaving only man-made lights, and then averaging data from all orbits of a given satellite in a given year to produce a satellite-year dataset. Each satellite-year dataset is a longitude-latitude grid reporting the intensity of light as a digital number (DN) between 0 and 63 for each 30 arcsecond output pixel. Henderson et al. analyzed 30 satellite-years but, for reasons that are not entirely clear,

the data from 2008 for satellite F15 are no longer available, so my replication relies on 29 satellite-years.[1]

Because of the curvature of the Earth, grid cell area varies with latitude; a pixel on the equator has an area of approximately 0.86 km^2, while a pixel on New York City has an area of approximately 0.65 km^2. Following the original paper, I account for this variation by computing the weighted average of lights across pixels in a country, weighting each pixel by its share of the country's land area. Land area is computed using Version 1 of the GRUMP data set (CIESIN 2011), and country boundaries are based on Version 3 of the GPW data set (CIESIN 2005). The original paper relied on an earlier version of GRUMP, which may introduce slight differences between my analysis and theirs.

Analysis

```
Joining with `by = join_by(iso)`
```

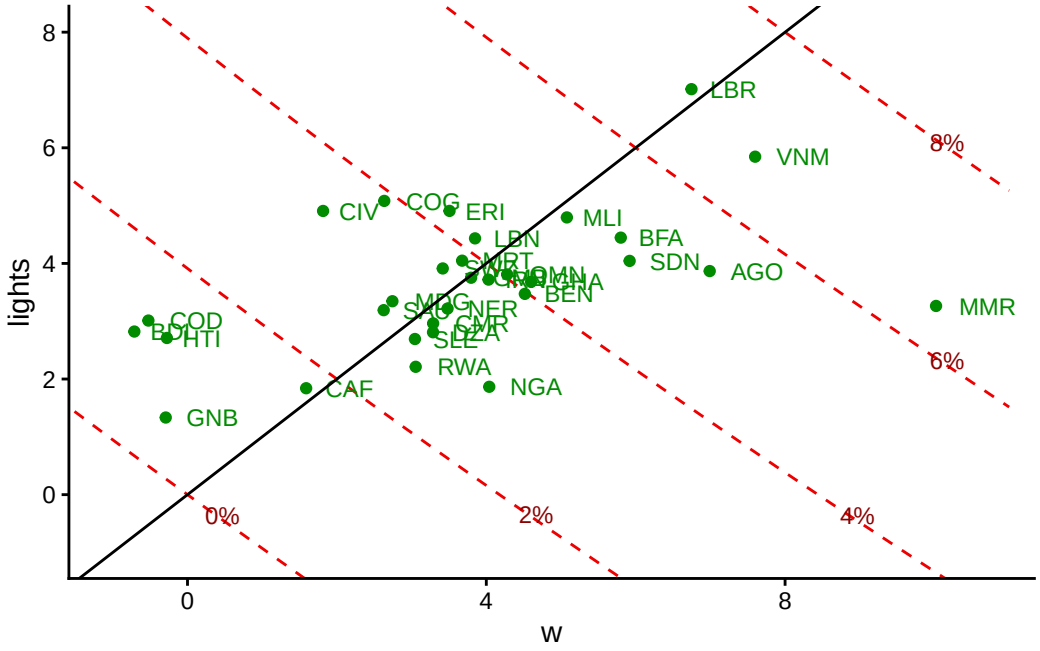


Figure 1: Growth in Fitted Lights vs. WDI GDP (LCU) for Bad Data Countries, 1992/93-2005/06

[1]In 2019, the archive hosting the DMSP-OLS data moved from NGDC to the Payne Institute for Public Policy at the Colorado School of Mines, and the 2008 F15 files may have been lost in the shuffle. I was unable to locate another source for the file.

Table 1: Baseline Results, 1992-2008

	ln(GDP)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ln(lights/area)	0.281***	0.270***	0.272***	0.288***	0.283***		0.168***
	(0.031)	(0.035)	(0.032)	(0.034)	(0.047)		(0.051)
ln(top-coded + 1)			0.011*				
			(0.006)				
ln(unlit + 1)			−0.006				
			(0.018)				
Spatial Gini				0.127			
				(0.212)			
ln(KWH)						0.283***	0.201***
						(0.047)	(0.040)
Observations	3016	3016	3016	3016	1853	1853	1853
Countries	188	188	188	188	128	128	128
(Within) R^2	0.770	0.771	0.772	0.771	0.758	0.767	0.782

* p <0.1, ** p <0.05, *** p <0.01
Clustered (country) standard errors in parentheses.

Table 2: Lights Up/Down, Time Trend, and Long Differences

	Fixed effects	Country time trend	Demeaned plus/minus	Long difference	
	(1)	(2)	(3)	(4)	(5)
ln(lights/area)	0.281***	0.193***		0.321***	0.304***
	(0.031)	(0.037)		(0.037)	(0.037)
$\lvert + \Delta$ ln(lights/area)$\rvert$			0.273***		
			(0.039)		
$\lvert - \Delta$ ln(lights/area)$\rvert$			−0.288***		
			(0.056)		
ln(top-coded + 1)					0.020
					(0.015)
ln(unlit + 1)					−0.010
					(0.033)
Time effects	Yes	Yes	In demean	No	No
Country effects	Yes	Yes	In demean	No	No
Observations	3016	3016	3016	170	170
Countries	188	188	188	170	170
(Within) R^2	0.770	0.201	0.214	0.276	0.284

* p <0.1, ** p <0.05, *** p <0.01
Clustered (country) standard errors in parentheses.

Table 3: Results for World Bank-Rated Countries

	Fixed effects	Country time trend	Long difference
Panel A			
ln(lights/area)	0.3096***	0.2795***	0.3284***
	(0.0382)	(0.0455)	(0.0462)
Constant			0.3652***
			(0.0281)
Observations	1954	1954	113
Countries	118	118	113
(Within) R^2	0.781	0.904	0.297
Panel B			
Difference in ψ for good data countries	0.0499	-0.0053	0.0916
	(0.0632)	(0.0656)	(0.0913)
Squared residuals on good data dummy	-0.0055***	-0.0017*	-0.0281
	(0.0017)	(0.0010)	(0.0183)

* p <0.1, ** p <0.05, *** p <0.01

Table 4: Calibrating the Statistical Model

Signal to total variance of measured income			Weight for measured income growth in calculation of true growth	
Good data countries: ϕ_g	Bad data countries: ϕ_b	Structural effect of true income growth on lights growth β	Good data countries: λ_g	Bad data countries: λ_b
1	0.660	1.021	1.000	0.563
0.9	0.594	1.134	0.850	0.479
0.8	0.528	1.276	0.699	0.394
0.7	0.462	1.458	0.549	0.309
0.6	0.396	1.701	0.398	0.224

30 bad data countries, 83 good data countries.

Table 5: Average Annual Growth Rates for Bad Data Countries, 1992/93-2005/06

Country	ISO	WDI GDP (LCU)	Optimal combination of fitted lights and WDI GDP	Difference
Myanmar	MMR	10.02	6.73	-2.99
Angola	AGO	6.99	5.48	-1.41
Nigeria	NGA	4.04	2.99	-1.01
Sudan	SDN	5.92	5.02	-0.85
Viet Nam	VNM	7.60	6.76	-0.78
Burkina Faso	BFA	5.80	5.15	-0.61
Benin	BEN	4.52	4.02	-0.48
Ghana	GHA	4.60	4.16	-0.42
Rwanda	RWA	3.06	2.65	-0.39
Algeria	DZA	3.29	3.06	-0.22
Oman	OMN	4.28	4.05	-0.22
Sierra Leone	SLE	3.04	2.88	-0.16
Cameroon	CMR	3.29	3.13	-0.15
Iran, Islamic Rep.	IRN	4.03	3.88	-0.14
Mali	MLI	5.08	4.94	-0.13
Niger	NER	3.48	3.36	-0.12
Gambia, The	GMB	3.80	3.78	-0.02
Central African Republic	CAF	1.59	1.71	0.12
Liberia	LBR	6.75	6.87	0.12
Mauritania	MRT	3.68	3.85	0.17
Eswatini	SWZ	3.42	3.65	0.23
Saudi Arabia	SAU	2.63	2.90	0.26
Lebanon	LBN	3.85	4.13	0.27
Madagascar	MDG	2.74	3.03	0.28
Eritrea	ERI	3.51	4.18	0.65
Guinea-Bissau	GNB	-0.29	0.48	0.78
Congo, Rep.	COG	2.63	3.80	1.13
Haiti	HTI	-0.28	1.14	1.42
Cote d'Ivoire	CIV	1.82	3.28	1.44
Congo, Dem. Rep.	COD	-0.52	1.15	1.69
Burundi	BDI	-0.71	0.96	1.69

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